Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	7 October 2026
Team ID	
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Script-Controlled ACL – Restrict Record Access Based on Field Value

Reference:
https://www.servicenow.com/docs/r/oJu9n0q6rU~F9UHAnNCt5w/aGZqCO71IpYGqS0MTeJC7Q

Technical Architecture – Script-Controlled ACL on ServiceNow

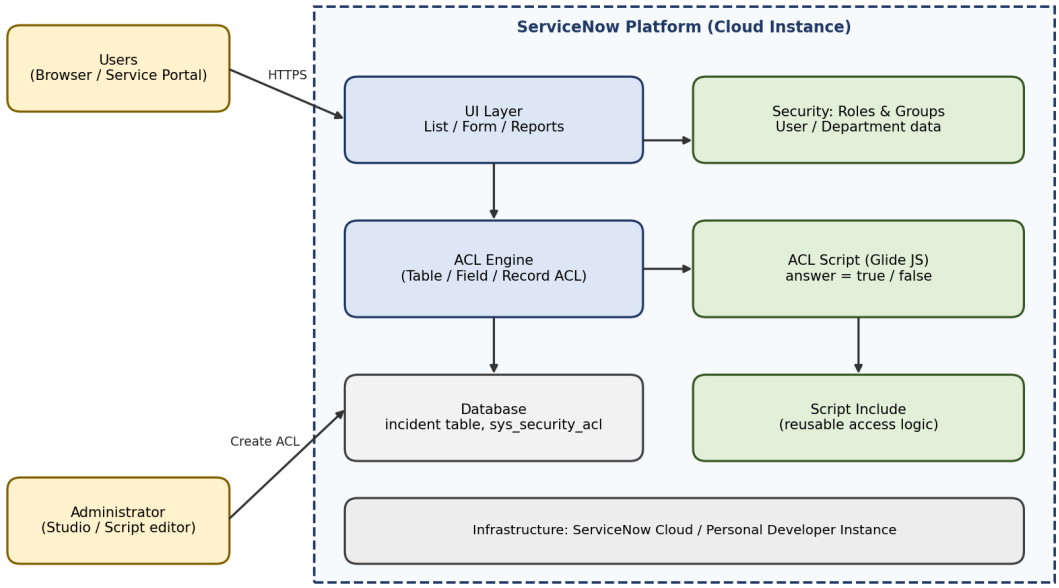


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	How the user opens records: list view, form view and reports	ServiceNow UI (Next Experience), Service Portal
2.	Application Logic-1	ACL script that checks the field value and the logged-in user (answer = true / false)	JavaScript (Glide API), ServiceNow ACL
3.	Application Logic-2	Reusable access logic called from the ACL	Script Include (JavaScript)

S.No	Component	Description	Technology
		script	
4.	Application Logic-3	Role and group check for privileged users	gs.hasRole(), User Groups
5.	Database	Records table and ACL rules storage	ServiceNow tables: incident (example), sys_security_acl
6.	Cloud Database	Database service provided with the instance	ServiceNow cloud instance database
7.	File Storage	Not required for this project (record attachments follow the same table ACL)	Not applicable
8.	External API-1	Not required for the core ACL logic	Not applicable
9.	External API-2	Optional: verify access through the REST Table API during testing	ServiceNow REST API
10.	Machine Learning Model	Not required for this project	Not applicable
11.	Infrastructure (Server / Cloud)	Application development and testing on a ServiceNow instance	ServiceNow Personal Developer Instance (Cloud)

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	The solution uses the built-in ServiceNow platform; no open-source framework is required	Not applicable
2.	Security Implementations	Record-level access control using roles and a script condition; deny unless the script returns true	ServiceNow ACL, Roles, Groups
3.	Scalable Architecture	The same Script Include and ACL pattern can be reused for other tables and field values	Script Include, ACL
4.	Availability	Rule runs inside the platform, so availability follows the ServiceNow instance	ServiceNow Cloud
5.	Performance	Keep the ACL script light: no heavy GlideRecord queries, check only the required fields	Optimised Glide scripting

References:

https://www.servicenow.com/docs/r/yokohama/platform-security/access-control/exploring-access-control-list.html?contentId=oczJQGZK~1lGzH~ITrwZHg

https://www.servicenow.com/docs/r/oJu9n0q6rU~F9UHAnNCt5w/aGZqCO71IpYGqS0MTeJC7Q

https://www.servicenow.com/docs/r/application-development/servicenow-sdk/acl-api-now-ts.html